

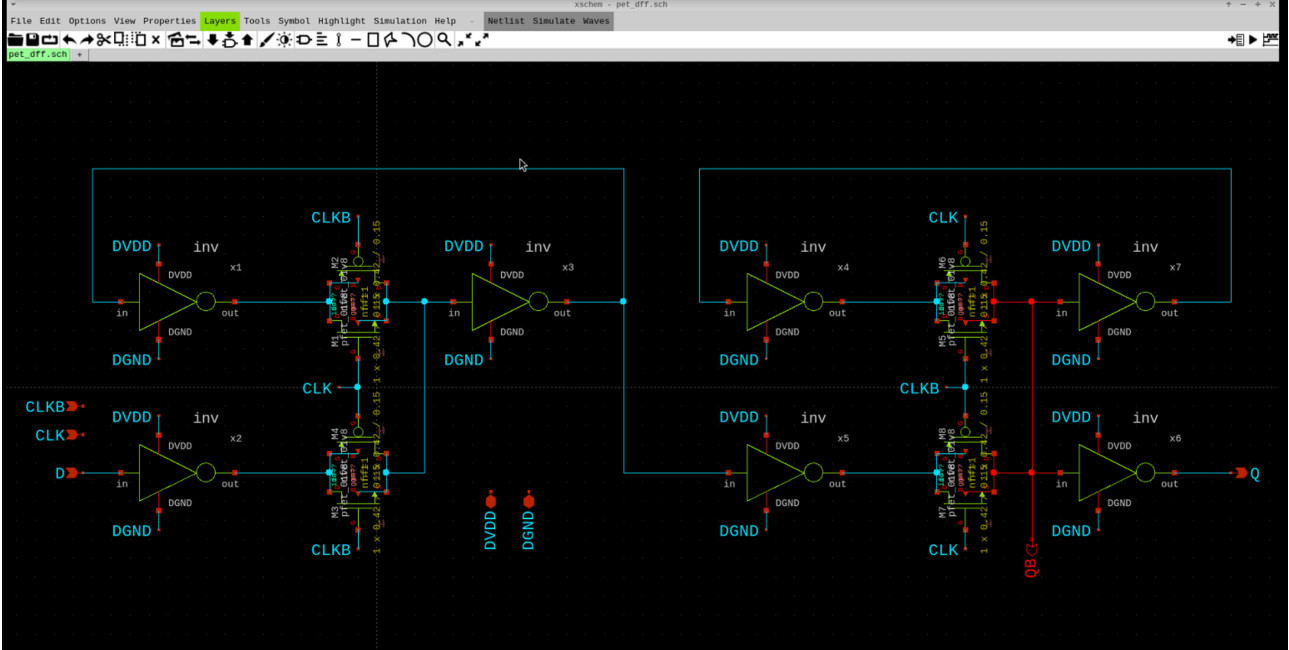
EE5311 Tutorial 6 Report

Name: Ishan Aji Hassan Roll: EE23B097

Question 1

Schematic

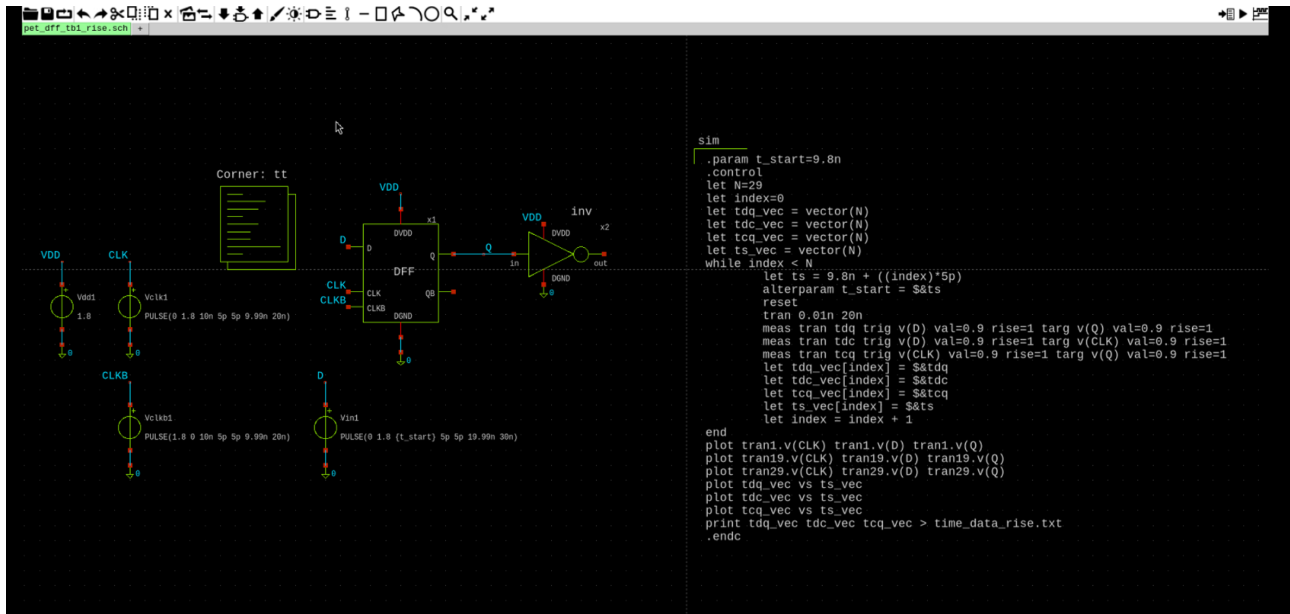
Positive edge-triggered D flip-flop (1× inverters: 0.42/0.15 nMOS, 0.84/0.15 pMOS; transmission gates: 0.42/0.15):



For both parts, the D transition time t_s is swept in 5 ps steps. t_{DQ} , t_{DC} and t_{CQ} are measured at the 50% (0.9 V) points. The t_{DC} and t_{CQ} at minimum t_{DQ} give t_{setup} and t_{CQ} .

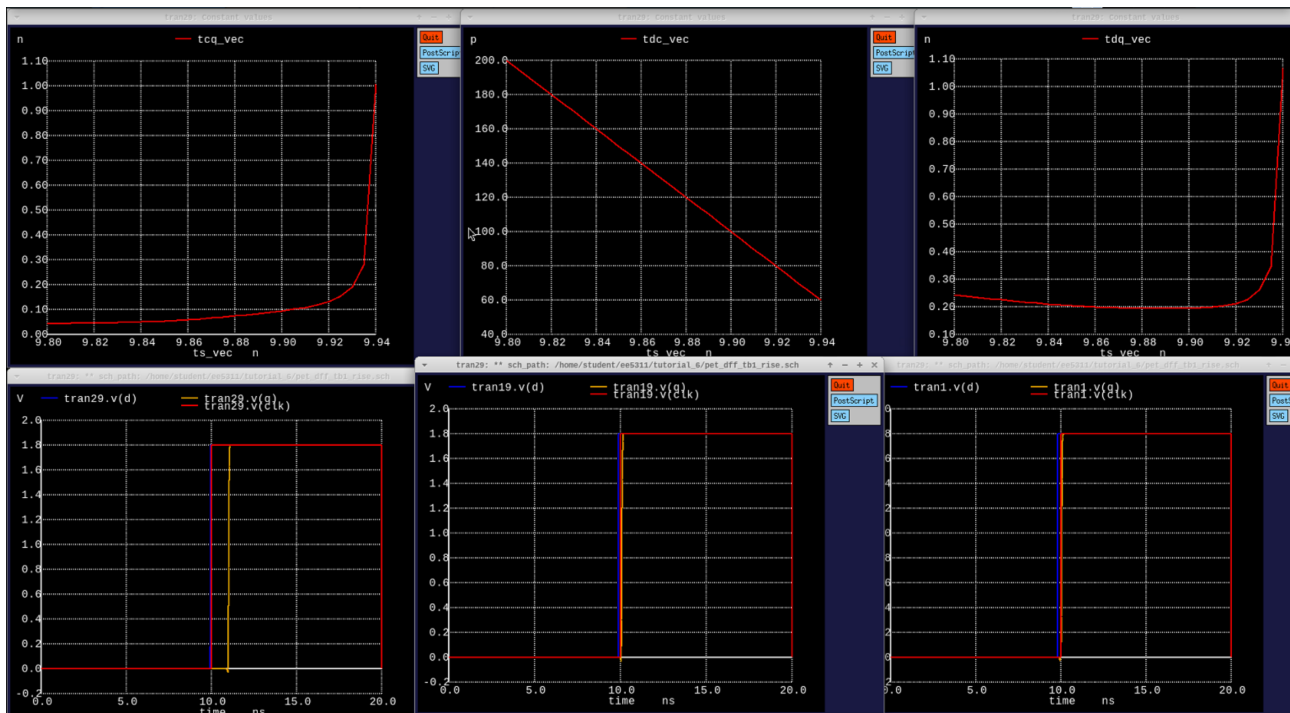
Part (a): Rising Q

Testbench



Measurement

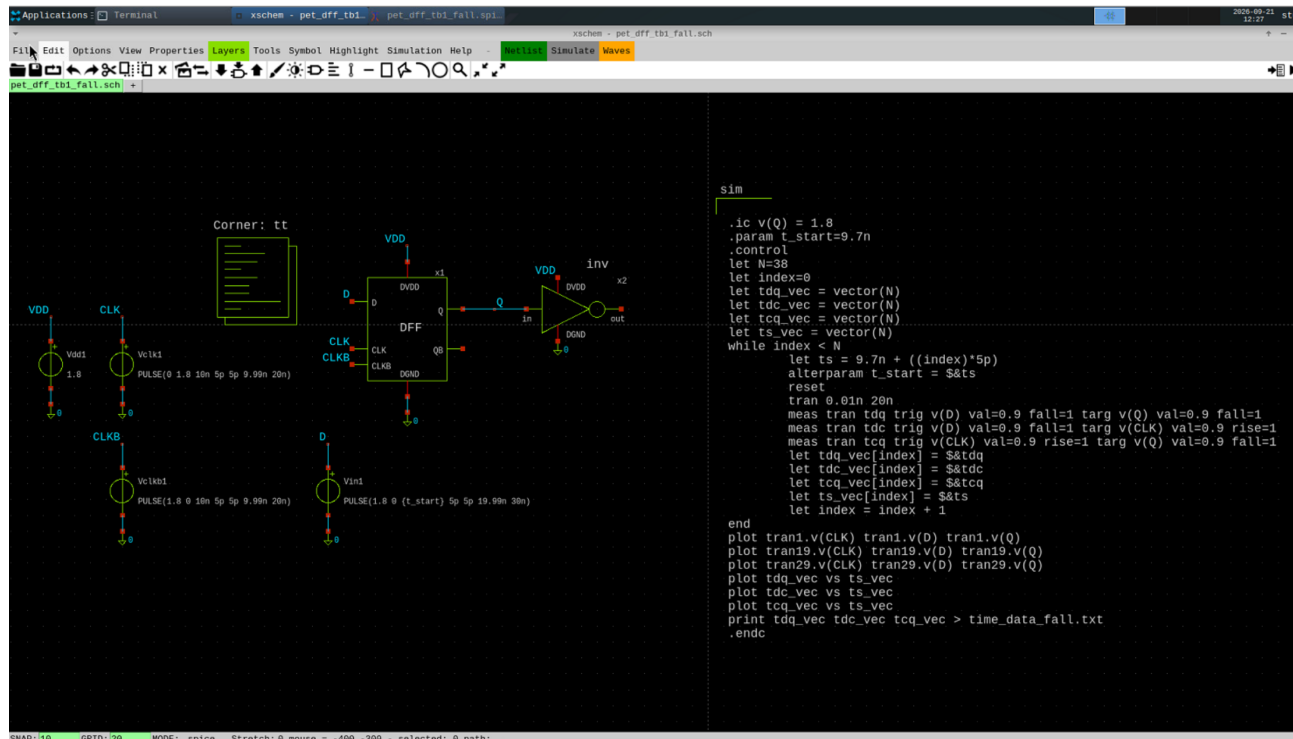
Top: t_{CQ} , t_{DC} , t_{DQ} vs. t_s ($t_s = 9.80\text{--}9.94\text{ ns}$). Bottom: CLK, D, Q for sweep points 29, 19 and 1.



Part (b): Falling Q

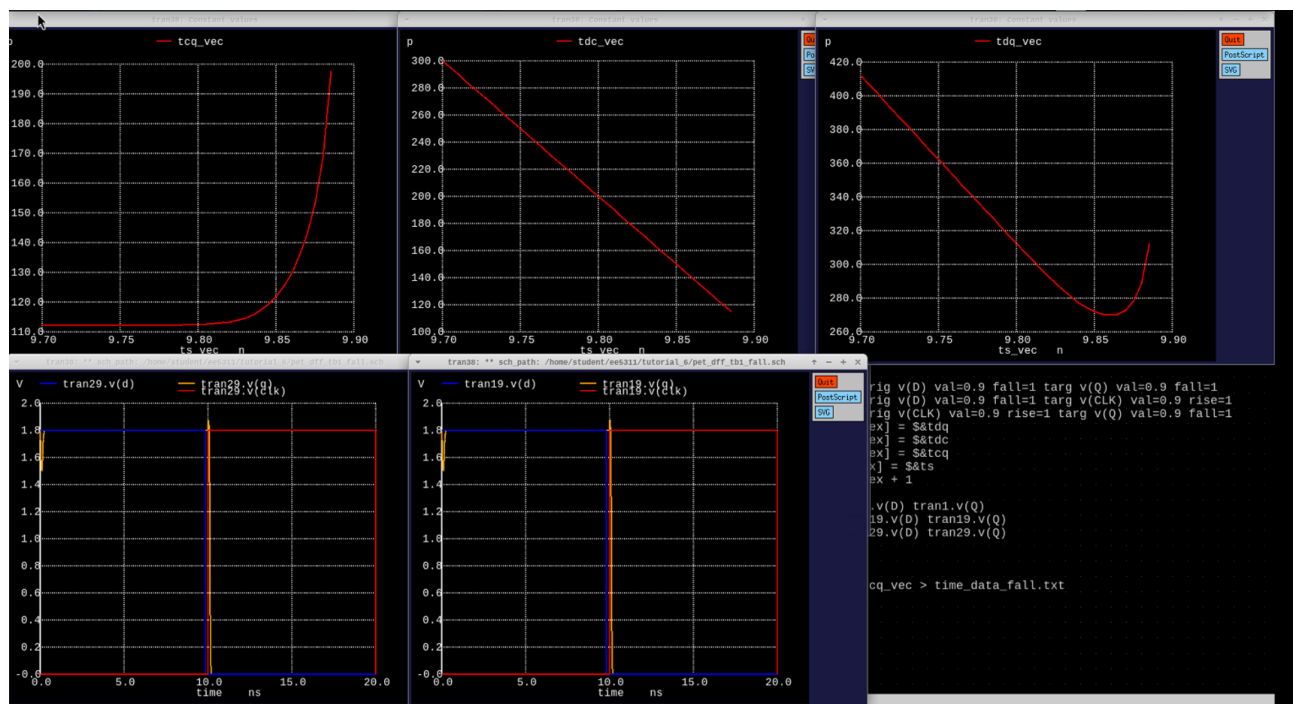
Testbench

The initial condition `.ic v(Q) = 1.8` sets the flip-flop to a defined high state before the falling transition.



Measurement

Top: t_{CQ} , t_{DC} , t_{DQ} vs. t_s ($t_s = 9.70\text{--}9.885\text{ ns}$). Bottom: CLK, D, Q for sweep points 29 and 19.



Finding minimum t_{DQ}

Python script run on the print output files (time_data_rise.txt, time_data_fall.txt):

```
import numpy as np

def analyse(fname, ts0):
    # columns: index, tdq, tdc, tcq (skip header lines)
    d = np.loadtxt(fname, skiprows=5)
    idx, tdq, tdc, tcq = d[:, 0], d[:, 1], d[:, 2], d[:, 3]
    i = np.argmin(tdq)
    ts = ts0 + i * 5e-12
    print(f"{fname}: min tDQ = {tdq[i]*1e12:.3f} ps at index {int(idx[i])} (ts = {ts*1e9:.3f} ns)")
    print(f"   tsetup = tDC = {tdc[i]*1e12:.1f} ps, tCQ = {tcq[i]*1e12:.3f} ps")

analyse("time_data_rise.txt", 9.80e-9)
analyse("time_data_fall.txt", 9.70e-9)
```

Output:

```
student@f3cb7f30520e:~/ee5311/tutorial_6$ python3 find_min.py
time_data_rise.txt: min tDQ = 194.311 ps at index 18 (ts = 9.890 ns)
   tsetup = tDC = 110.0 ps, tCQ = 84.311 ps
time_data_fall.txt: min tDQ = 270.016 ps at index 32 (ts = 9.860 ns)
   tsetup = tDC = 140.0 ps, tCQ = 130.016 ps
student@f3cb7f30520e:~/ee5311/tutorial_6$
```

Results

Q transition	Min. index	t_s (ns)	t_{DQ} (ps)	$t_{\text{setup}} = t_{DC}$ (ps)	t_{CQ} (ps)
(a) Rising	18	9.890	194.311	110	84.311
(b) Falling	32	9.860	270.016	140	130.016

Rising Q: $t_{\text{setup}} = 0.11$ ns, $t_{CQ} = 0.0843$ ns.

Falling Q: $t_{\text{setup}} = 0.14$ ns, $t_{CQ} = 0.130$ ns.

The sweep step is 5 ps, so the values are accurate to about ± 5 ps.